Emergency responder radio coverage test: Example: public safety ERCES, 3-storey office (completed example)

Example office building, Arlington VA (fictional) · Example County Fire Marshal (example) · IFC 2021 510 + NFPA 1225 (2022) · prepared 2026-10-28

Result: PASS

Pass criteria: each test location ≥ -95 dBm downlink and uplink, DAQ ≥ 3 both ways (or SINR ≥ 18 dB with BER ≤ 2.5%), on every channel tested there (the worst channel is shown); 95% of general areas and 99% of critical areas on each floor. Each floor's grid: IFC 510.5.4: 20 areas per floor; more than 1 failed area fails the floor. With exactly 2 failed, the floor may be re-divided into 40 areas and then passes with no more than 2 failed areas, none sharing an edge with another; failing that, the system is altered to reach 95%. Critical squares and named critical areas also meet 99% (profile: IFC 2021 (refs NFPA 1221): –95 dBm + DAQ 3.0). Only readings taken with an instrument calibrated within 12 months of the reading are scored.

Building and people

Building	Example: public safety ERCES, 3-storey office (completed example) · Example office building, Arlington VA (fictional) · 3 floors · 72,000 sq ft · Thermally efficient (low-E glass, foil insulation), typical (31 dB)
AHJ / fire marshal	Dana Whitfield · Example County Fire Marshal's Office (example) · 703-555-0142 · dwhitfield@firemarshal.example
Building owner	Priya Raman · Sample Properties LLC (example) · 703-555-0117 · praman@sampleproperties.example
Radio system owner	Marcus Bell · Example County ECC radio shop (example) · 703-555-0163 · radioshop@county-ecc.example
General contractor	Tom Alvarez · Sample Builders Inc. (example) · 703-555-0125 · talvarez@samplebuilders.example
Subcontractor	Lena Ortiz · Sample Electric Co. (example) · 703-555-0138 · lortiz@sampleelectric.example
Our technician	Jordan Lee · Sample RF Services (example) · 703-555-0109 · jlee@samplerf.example

Tester and instruments

Tester	Jordan Lee (example)
FCC GROL	PG-EX-0001 (example)
NICET	Level III (example)
Test dates	2026-10-20 to 2026-10-21

Instrument	Serial	Calibrated	Status
PCTEL IBflex	**EX-1001**	**2026-06-15**	current

Compliance

Licensee / agency consent to retransmit (47 CFR 90.219(b)(1))	2026-09-02
FCC signal booster registration (Class B: 47 CFR 90.219(d)(5))	2026-10-22
AHJ permit	2026-08-15

Design approved	2026-09-10
Monitoring connected to the fire alarm panel	2026-10-16
Acceptance test passed	2026-10-27
Battery test	2026-10-21
Annual retest due	2027-10-27
Compared with	the acceptance test of 2026-10-27
Material rules confirmed	22: UL 2524 listing of active equipment; Enclosures for BDAs and batteries; Secondary (standby) power; Supervisory signals to the fire alarm system; Dedicated annunciator; Backbone survivability (fire resistance); Older-edition survivability rules (jurisdictions still on them); Cable fire ratings (plenum / riser); Mechanical protection of coax; Lightning / surge protection on the donor line; Grounding the coax shield at entry; Donor antenna mounting and isolation; Oscillation control; FCC Part 90 signal booster class, consent and registration; Frequencies and upgradeability; Near-far and antenna density; Link budget: choosing cable size; Component power ratings; Low-PIM choices for multi-carrier DAS; Radiating cable stop bands; Fiber connectors for RF-over-fiber DAS; Labelling and signage

Systems covered

Agency	Call sign	Downlink MHz	Uplink MHz	Type
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	851.7125	806.7125	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	851.8125	806.8125	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	852.675	807.675	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	853.1875	808.1875	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.4375	811.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.7625	811.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.9375	811.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.4375	812.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.7625	812.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.9375	812.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.4375	813.4375	Analog FM / Unknown

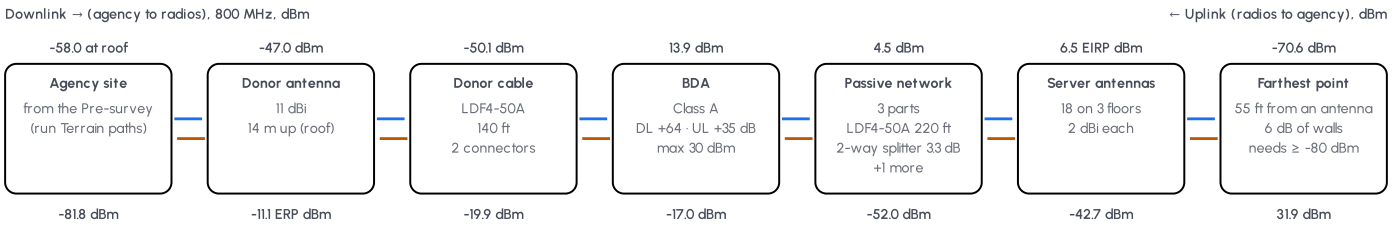
Agency	Call sign	Downlink MHz	Uplink MHz	Type
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.7625	813.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.9375	813.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.4375	814.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.7625	814.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.9375	814.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.4375	815.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.7625	815.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.9375	815.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KIC338	154.13	153.89	Analog FM
Arlington County Department of Public Safety Communications and Emergency Management	KIC338	154.28	153.89	Analog FM

Licensee consent (47 CFR 90.219(b)(1))

Licensee	Call signs	Consent	Date
Arlington County Department of Public Safety Communications and Emergency Management	KIC338, KNIQ704	given	2026-09-02
METROPOLITAN WASHINGTON AIRPORTS AUTHORITY	WPAY961, WPQB443, WQYK669	not needed	2026-09-02
City of Alexandria	KA9678, WNFS909, WQPF485, WQPG622	not needed	2026-09-02
MONTGOMERY, COUNTY OF	KCP601, WNPN638, WPBH837, WPWJ571, WPWL719, WREP612, WRHY973, WRHY978	not needed	2026-09-02
Radio Communications	WNHB839, WPVA690, WPVK244, WZX459	not needed	2026-09-02

Licensee	Call signs	Consent	Date
Prince George's County Government-OHS	KGA361, WNBR219, WQJS954, WQLP960, WQSH369, WRAN914, WRAN915, WRAP365, WRPB615	not needed	2026-09-02
Baltimore Gas and Electric Company	KNDB915, KNDH602, WQAP338, WQCK465, WQLC435, WQLY679	not needed	2026-09-02
DISTRICT OF COLUMBIA	KNJA391, KNJE616, KNJU834, WPXT459, WPYD604, WRBZ373	not needed	2026-09-02
WASHINGTON GAS LIGHT COMPANY	WRVQ768, WRWK328, WRWR905, WRXC754, WRXC755, WRXC757	not needed	2026-09-02
FAIRFAX, COUNTY OF	KNIH412, WNAJ365, WNYZ445, WNYZ447, WPPA560, WPVY861, WQCP394	not needed	2026-09-02
MONTGOMERY COUNTY, MARYLAND	WQZC785, WQZC983	not needed	2026-09-02
Fairfax County Public Schools	WRJI389	not needed	2026-09-02
State of Maryland - UNIVERSITY OF MARYLAND	WPTZ482	not needed	2026-09-02
Howard, County of MD	WPXJ396	not needed	2026-09-02

System diagram: signal path, start to finish



Levels are the 800 MHz link budget's worst case (860.9375 MHz down, 815.9375 MHz up); every band is in the table below. Blue: what reaches the radios inside; orange: what reaches the agency's receiver.

Band	Systems	At roof dBm	BDA out dBm	Antenna EIRP dBm	Farthest point dBm	Needs dBm	Uplink at site dBm	Checks
800 MHz	Arlington County Department of Public Safety Communications and Emergency	-58.0	13.9	6.5	-70.6	-80.0	-81.8	pass

Band	Systems	At roof dBm	BDA out dBm	Antenna EIRP dBm	Farthest point dBm	Needs dBm	Uplink at site dBm	Checks
	Management							
VHF High	Arlington County Department of Public Safety Communications and Emergency Management	-60.0	15.6	8.9	-51.2	-80.0	-76.9	pass

No donor paths yet: run Terrain paths on the Pre-survey to add each agency site's distance, bearing and terrain.

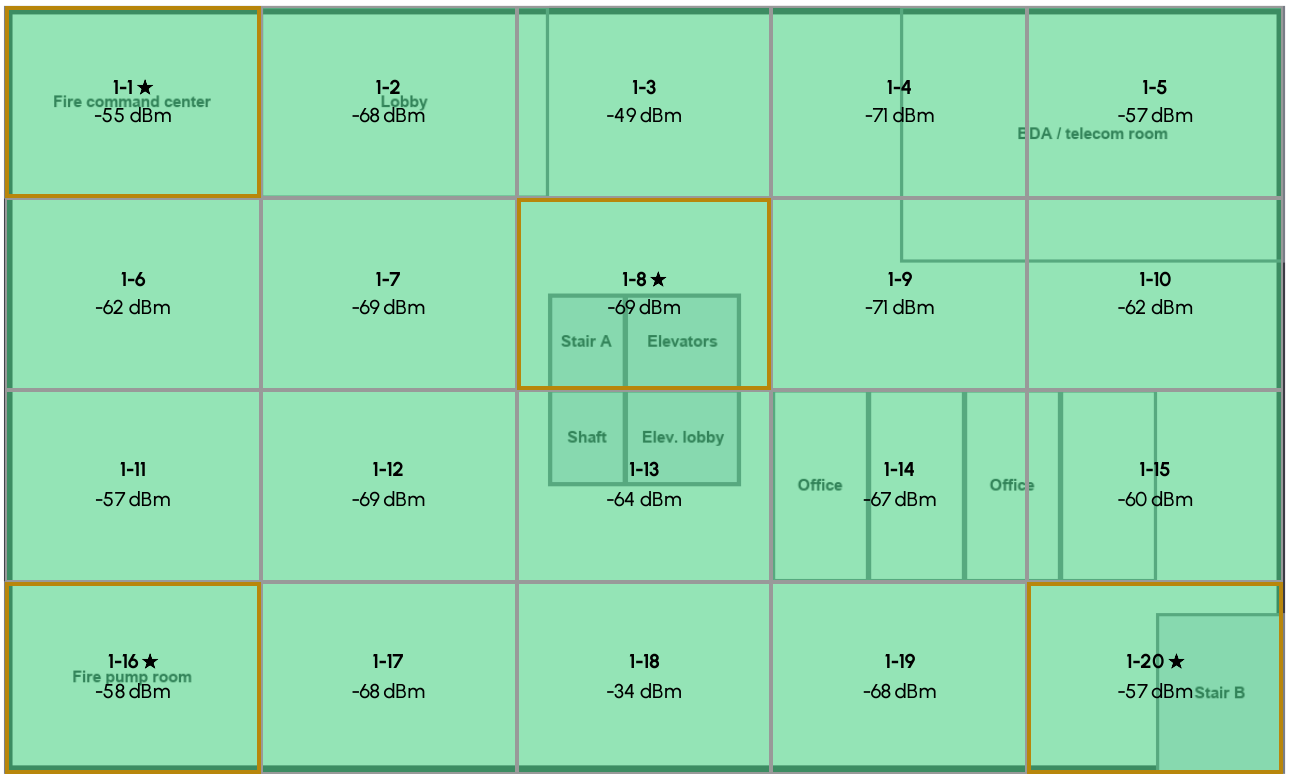
Antennas by floor: Floor 1: 6 · Floor 2: 6 · Floor 3: 6

Summary by floor

Floor	General pass	Critical pass	Untested	Not counted	Result	Rule and reason
1	16/16 (100%)	4/4 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
2	18/18 (100%)	2/2 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
3	18/18 (100%)	2/2 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
Named critical areas		5/5	0			each must pass

Floor 1: 16/16 general, 4/4 critical pass · PASS

IFC 510.5.4, 20 areas: No area failed.



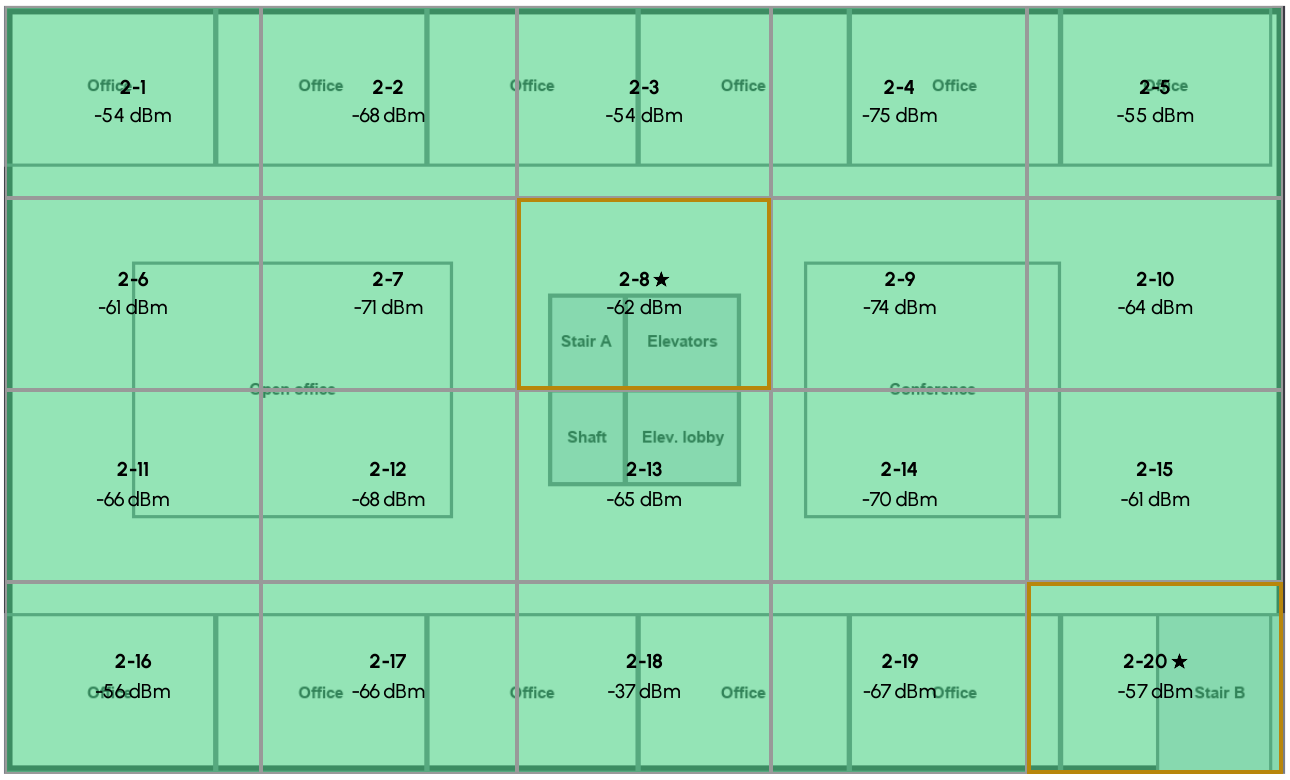
Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
1-1	★	154.13 (worst of 2)	**−55**	**4.6**		**−62**	**4.6**	**+0 dB**	PCTEL IBflex EX-1001	pass	
1-2		154.13 (worst of 2)	**−68**	**4.4**		**−79**	**4.2**	**+0 dB**	PCTEL IBflex EX-1001	pass	
1-3		851.7125 (worst of 2)	**−49**	**4.6**		**−59**	**4.3**	**+0 dB**	PCTEL IBflex EX-1001	pass	
1-4		851.7125 (worst of 2)	**−71**	**4.2**		**−76**	**4**	**+0 dB**	PCTEL IBflex EX-1001	pass	
1-5		154.13 (worst of 2)	**−57**	**4.4**		**−67**	**4.4**	**+0 dB**	PCTEL IBflex EX-1001	pass	

Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
1-6		851.7125 (worst of 2)	−62	4.2		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-7		851.7125 (worst of 2)	−69	4.1		−73	3.8	+0 dB	PCTEL IBflex EX-1001	pass	
1-8	★	851.7125 (worst of 2)	−69	4.4		−81	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
1-9		851.7125 (worst of 2)	−71	4.2		−83	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
1-10		851.7125 (worst of 2)	−62	4.3		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-11		154.13 (worst of 2)	−57	4.3		−66	3.9	+0 dB	PCTEL IBflex EX-1001	pass	
1-12		851.7125 (worst of 2)	−69	4.2		−80	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-13		154.13 (worst of 2)	−64	4.4		−73	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
1-14		154.13 (worst of 2)	−67	4.1		−81	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-15		154.13 (worst of 2)	−60	4.3		−70	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
1-16	★	851.7125 (worst of 2)	−58	4.4		−68	4.1	+0 dB	PCTEL IBflex EX-1001	pass	Fire pump room: pump running, door closed; the fire marshal witnessed this spot (example).
1-17		851.7125 (worst of 2)	−68	4.6		−77	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
1-18		851.7125 (worst of 2)	−34	4.6		−42	4.7	+0 dB	PCTEL IBflex EX-1001	pass	

Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
1-19		154.13 (worst of 2)	**−68**	**4.4**		**−78**	**4.4**	**+0 dB**	PCTEL IBflex EX-1001	pass	
1-20	★	851.7125 (worst of 2)	**−57**	**4.6**		**−66**	**4.5**	**+0 dB**	PCTEL IBflex EX-1001	pass	

Floor 2: 18/18 general, 2/2 critical pass · PASS

IFC 510.5.4, 20 areas: No area failed.

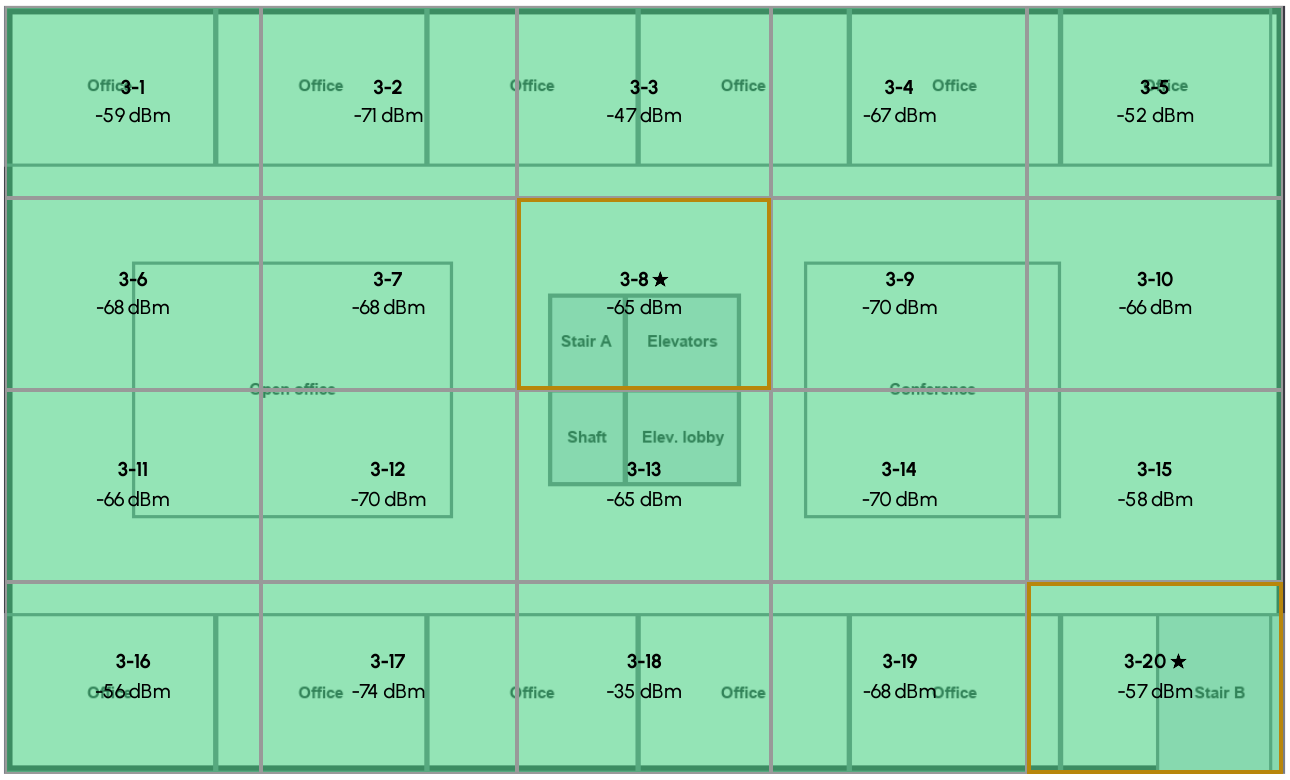


Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
2-1		851.7125 (worst of 2)	−54	4.5		−58	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-2		154.13 (worst of 2)	−68	4.2		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-3		851.7125 (worst of 2)	−54	4.5		−67	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
2-4		851.7125 (worst of 2)	−75	4.1		−79	4	+0 dB	PCTEL IBflex EX-1001	pass	
2-5		154.13 (worst of 2)	−55	4.2		−66	4	+0 dB	PCTEL IBflex EX-1001	pass	
2-6		154.13 (worst of 2)	−61	4.4		−72	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-7		154.13 (worst of 2)	−71	4.3		−76	4.1	+0 dB	PCTEL IBflex EX-1001	pass	

Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
2-8	★	154.13 (worst of 2)	−62	4.3		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-9		851.7125 (worst of 2)	−74	4		−82	3.7	+0 dB	PCTEL IBflex EX-1001	pass	
2-10		851.7125 (worst of 2)	−64	4.2		−76	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-11		851.7125 (worst of 2)	−66	4.3		−76	4	+0 dB	PCTEL IBflex EX-1001	pass	
2-12		851.7125 (worst of 2)	−68	4.3		−77	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-13		851.7125 (worst of 2)	−65	4.6		−73	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
2-14		851.7125 (worst of 2)	−70	4.1		−72	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-15		851.7125 (worst of 2)	−61	4.5		−67	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
2-16		154.13 (worst of 2)	−56	4.6		−69	4.7	+0 dB	PCTEL IBflex EX-1001	pass	
2-17		851.7125 (worst of 2)	−66	4.2		−77	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-18		851.7125 (worst of 2)	−37	4.6		−48	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
2-19		851.7125 (worst of 2)	−67	4.3		−75	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-20	★	154.13 (worst of 2)	−57	4.6		−60	4.4	+0 dB	PCTEL IBflex EX-1001	pass	

Floor 3: 18/18 general, 2/2 critical pass · PASS

IFC 510.5.4, 20 areas: No area failed.



Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
3-1		851.7125 (worst of 2)	−59	4.6		−67	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
3-2		851.7125 (worst of 2)	−71	4.2		−79	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-3		851.7125 (worst of 2)	−47	4.5		−52	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
3-4		154.13 (worst of 2)	−67	4.5		−70	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
3-5		851.7125 (worst of 2)	−52	4.4		−59	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-6		154.13 (worst of 2)	−68	4.1		−77	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-7		851.7125 (worst of 2)	−68	4.3		−82	4	+0 dB	PCTEL IBflex EX-1001	pass	

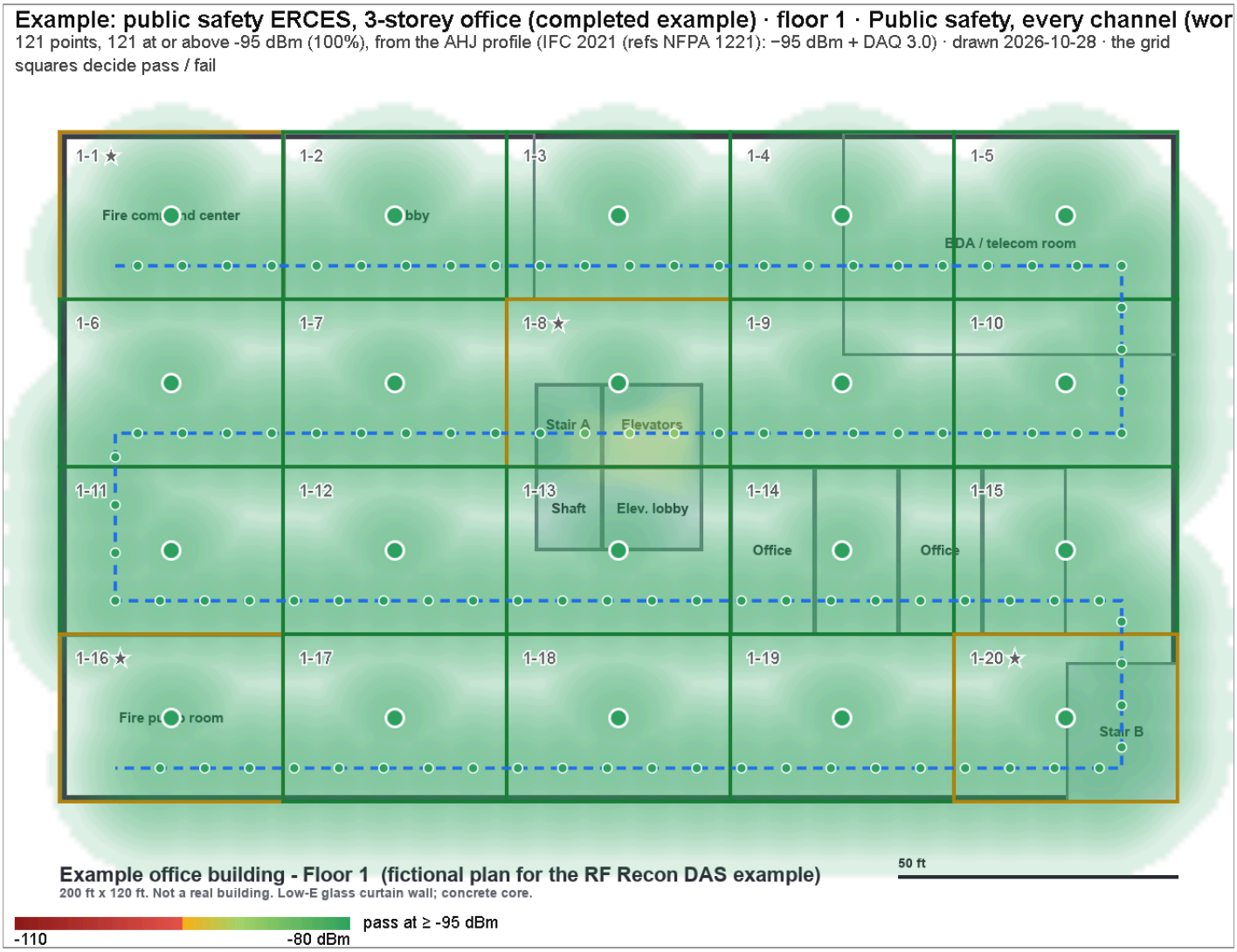
Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
3-8	★	851.7125 (worst of 2)	−65	4.3		−70	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-9		154.13 (worst of 2)	−70	4.3		−80	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-10		154.13 (worst of 2)	−66	4.5		−74	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
3-11		851.7125 (worst of 2)	−66	4.5		−74	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-12		851.7125 (worst of 2)	−70	4.3		−82	3.9	+0 dB	PCTEL IBflex EX-1001	pass	
3-13		851.7125 (worst of 2)	−65	4.6		−73	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
3-14		851.7125 (worst of 2)	−70	4.3		−73	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
3-15		851.7125 (worst of 2)	−58	4.5		−60	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-16		154.13 (worst of 2)	−56	4.4		−61	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-17		154.13 (worst of 2)	−74	4		−86	3.8	+0 dB	PCTEL IBflex EX-1001	pass	
3-18		154.13 (worst of 2)	−35	4.4		−40	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-19		154.13 (worst of 2)	−68	4.3		−78	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-20	★	851.7125 (worst of 2)	−57	4.6		−62	4.6	+0 dB	PCTEL IBflex EX-1001	pass	

Named critical areas

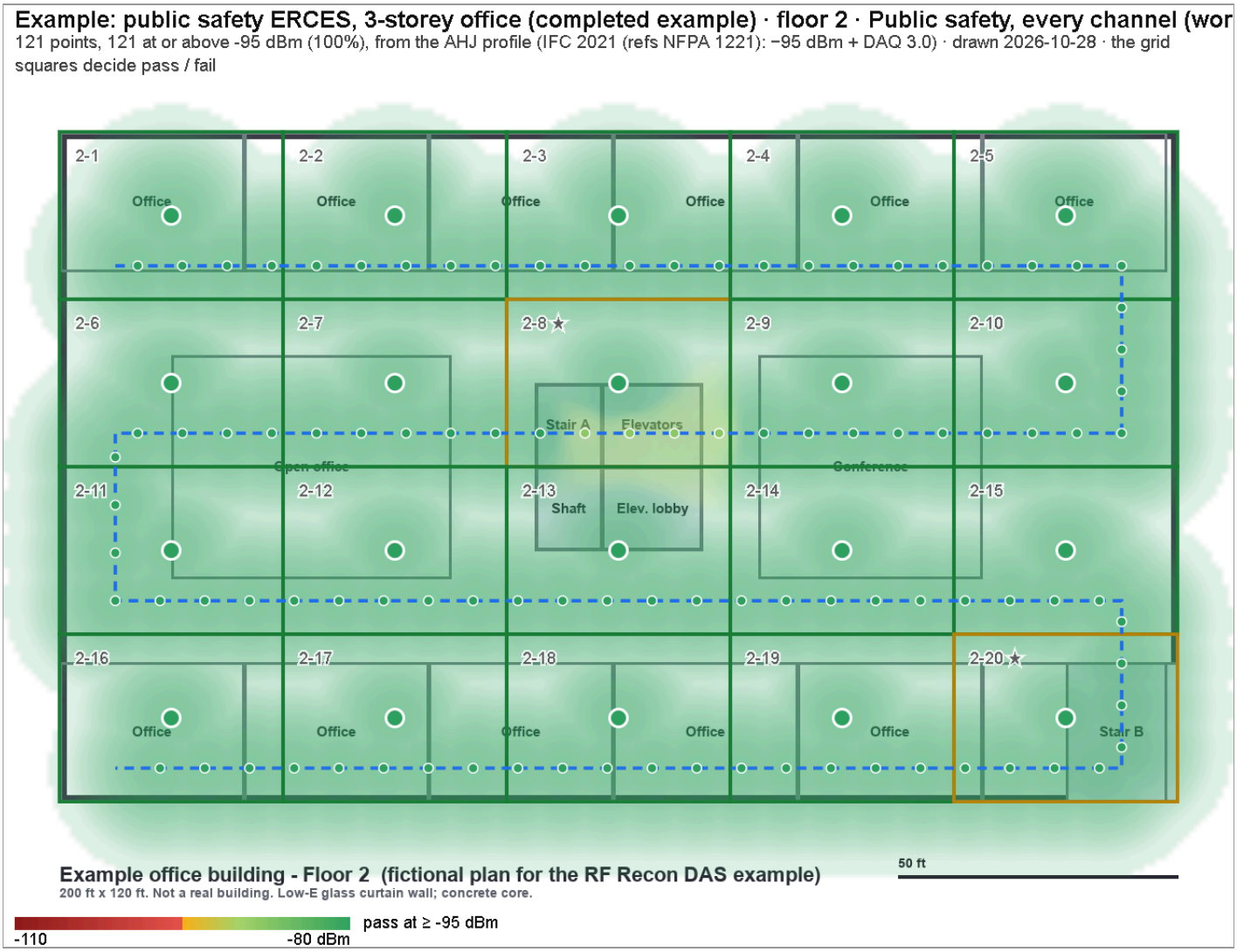
Area	DL dBm	DL DAQ	UL dBm	UL DAQ	Instrument	Result
Fire command center	−71	4.4	−75	4.1	PCTEL IBflex EX-1001	pass
Fire pump room	−84	3.8	−86	3.6	PCTEL IBflex EX-1001	pass
Exit stairs	−82	3.9	−84	3.7	PCTEL IBflex EX-1001	pass
Elevator lobbies	−77	4.2	−80	4	PCTEL IBflex EX-1001	pass
BDA / telecom room	−69	4.5	−72	4.3	PCTEL IBflex EX-1001	pass

Walk test heat maps

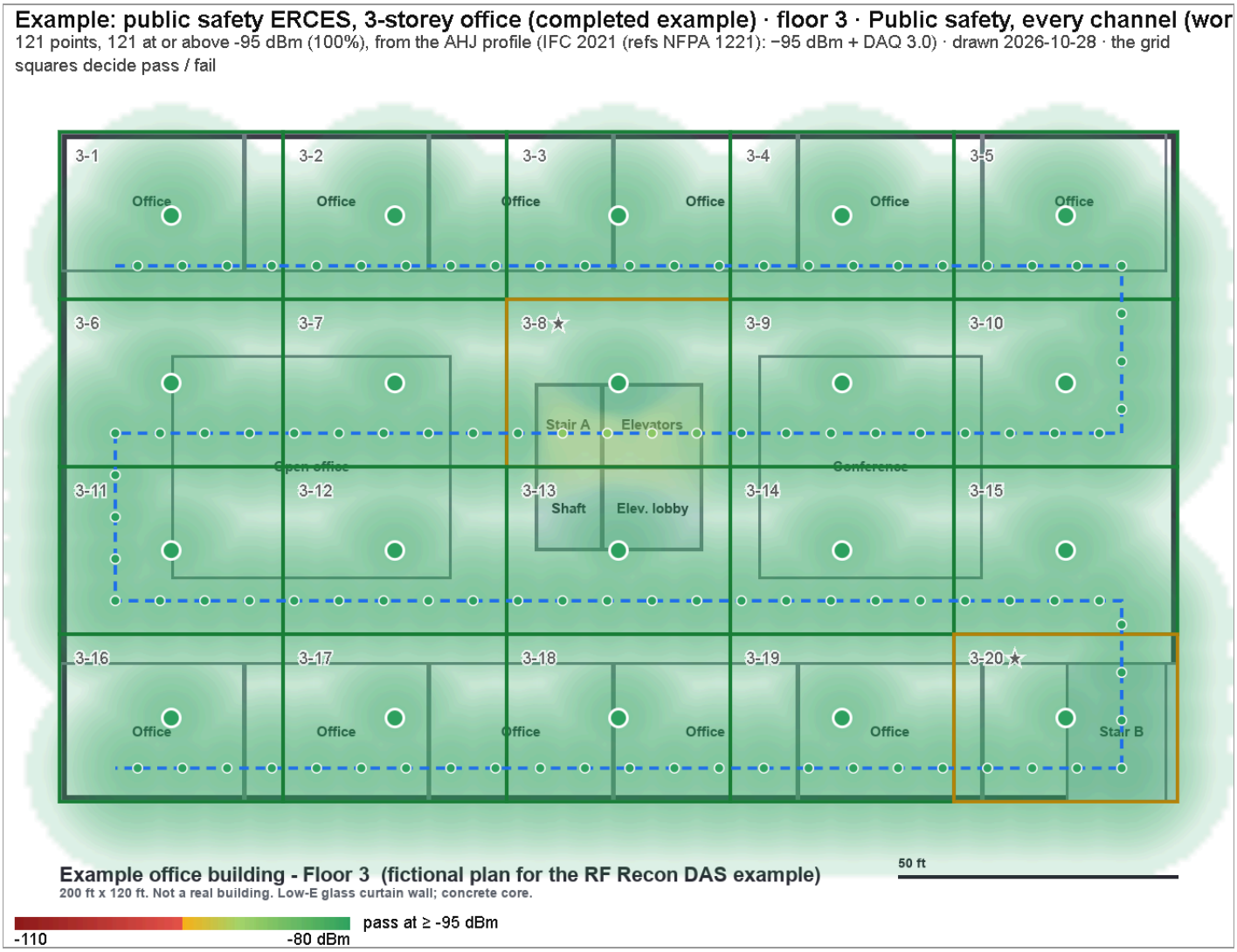
Readings walked on each floor's plan, coloured against the requirement. The heat map shows every reading, calibrated or not; the grid squares above decide pass / fail.



Floor 1 · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail



Floor 2 · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail



Floor 3 · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail

This product uses the FCC Data API but is not endorsed or certified by the FCC.